

P1 Architecture Roadmap

Disputes & Chargebacks Program Artifact

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P0	Implement canonical lifecycle, policy guardrails, and stage-level evidence.
P1	Scale to modular services for decisioning, compliance, and ledger integrity.
Metrics	Improve win rates, reduce net loss, and preserve host/guest trust KPIs.

P1 Architecture Roadmap (2-6 Weeks)

Objective

Evolve from demo UI into scalable disputes platform architecture supporting new payment methods, geographies, and verticals.

Target Architecture

1. Domain Services

- Dispute Lifecycle Service
 - Owns state machine and transition authorization.
- Decisioning Service
 - Rule-based + ML score blending.
 - Outputs recommendation, confidence, and expected value.
- Policy Compliance Service
 - Encodes deadlines and eligibility constraints.
 - Provides deterministic pass/fail reasons.
- Ledger Service
 - Immutable financial events + balance projections.
- Evidence Service
 - Evidence package completeness scoring.
 - Metadata validation and anti-fraud checks.

2. Data Model (Core Entities)

- dispute_case
- dispute_stage_event
- monetary_ledger_event
- evidence_item
- evidence_requirement

- policy_evaluation
- decision_recommendation
- reason_code_rule
- payment_method_rule
- risk_snapshot

3. Decisioning Stack

- Rules-first orchestration:
 - Hard policy constraints -> eligibility gating.
 - Reason-code/payment-method playbook.
 - ML risk/win scores as ranking/prioritization features.
 - Add explainability payload:
 - top_signals
 - missing_signals
 - confidence drivers

4. Eventing and Integrations

- Processor webhook ingestion boundary.
- Event bus for stage, evidence, and ledger updates.
- Retry-safe idempotency keys for chargeback events.
- Audit trail for policy-sensitive actions.

5. UX Architecture

- One shared case-expansion renderer with table sections:
 - Case Summary Table
 - Timeline Activity Table
 - Payment/Ledger Table
 - Evidence Package Table
 - Role overlays only (Customer/Merchant/Admin permissions), not separate logic forks.

Security and Integrity

- Fraud-resistance checks for doctored evidence and inconsistent metadata.
- Role-based permissioning for sensitive actions.
- Immutable financial event append-only model.
- SLA monitoring for response windows.

Observability

- Lifecycle funnel metrics:
- Ingested -> Deflected -> Represented -> Pre-Arb -> Closed
 - Win/loss by reason code, processor, payment method, region.
- Alerting:
- deadline at-risk
- failed clawback
- high-loss clusters

Rollout Plan

Phase 1

- Introduce shared domain modules behind existing UI.
- Add schema version and migration support.

Phase 2

- Move lifecycle + policy logic to service abstraction.
- Introduce immutable ledger events and reconciliation checks.

Phase 3

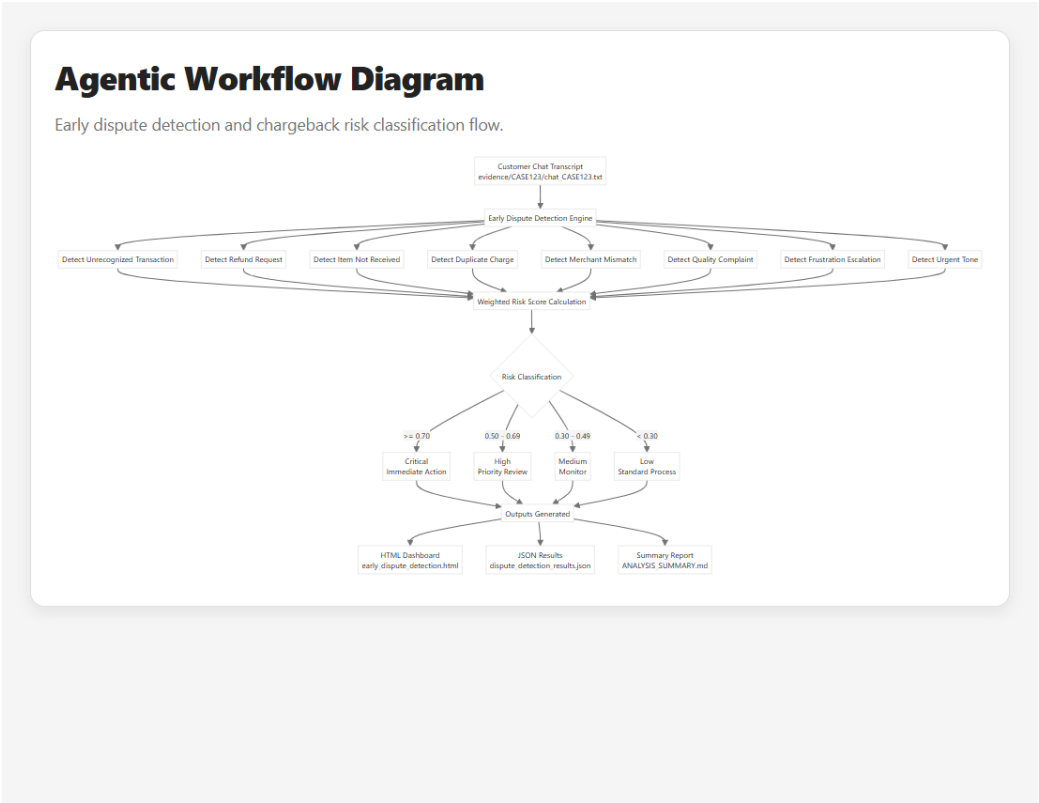
- Plug in score providers (ML + external risk vendors).
- Add experiment framework for deflection and outreach variants.

Success Criteria

- 0 ambiguous stage transitions.
- 100% explainable recommendations for operator actions.
- Reduced manual intervention in evidence collection.
- Lower net loss rate with stable trust and satisfaction indicators.

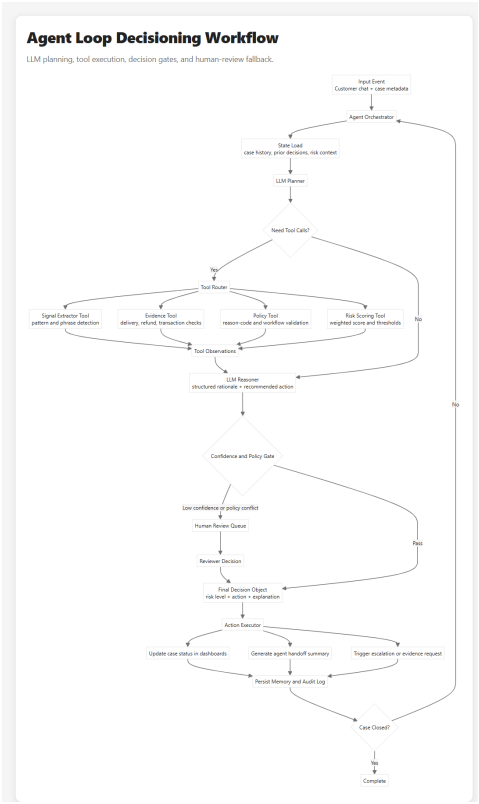
Visual Evidence Appendix

Agentic Workflow Diagram: early-dispute detection and classification flow



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Agent Loop Workflow: LLM planning, tools, and policy gate controls



File: 02_agent_loop_workflow.png

Merchant Risk Dashboard: operations queue, evidence intake, and actions

Merchant Risk & Operations Dashboard

Focused drilldown for the selected merchant, with unresolved case workflow, evidence actions, issuer simulation, and marketplace case creation support.

No active dispute cases were found for **Unknown Merchant**. Use the Admin Dashboard to navigate to another merchant.

Cases & Resolutions

Track dispute status and manage case resolutions. Click case ID to view chargeback lifecycle and outcome details.

+ Simulate Case

⌵ Simulate Status Change...

✓ Mark Evidence Ready

Open Cases

Resolved Cases

☐

Case ID

Amount

Reason Code

Initiated

Risk

Status

Days

Amount Due/Due

Fee

Payout

Evidence Documents

Manage and submit evidence documents to support dispute defense. Merchants can upload, review, and submit documents directly to the issuer.

Document Types Accepted: Transaction records, delivery proof, authorization logs, customer communications, merchant policies, payment receipts, shipping evidence, refund records, and fraud investigation reports.

Add New Evidence Document

Document Type

Transaction Record

Upload Document

Recommended Merchant Actions

Step 1: Confirm transaction details, capture payment authorization logs, and flag unusual routing or account changes.

Step 2: Prioritize high/critical cases for immediate evidence collection and escalate cases with issuer requests.

Step 3: Use the simulation tools at the top to practice issuer responses and bulk marketplace case actions.

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Customer Expanded Case: AI summary, money movement, and lifecycle reference

Resolution Center - Customer

Case ID

Amount

Reason Code

Initiated

Risk

Status

Days

Amount Due/Due

Fee

Payout

Case Details

Case Summary

Case History

Case Actions

Case Notes

Case Attachments

Case Comments

Case Status

Case Resolution

Case Outcome

Case Feedback

Case Rating

Case Review

Case Approval

Case Denial

Case Cancellation

Case Deletion

Case Archiving

Case Export

Case Import

Case Search

Case Filter

Case Sort

Case Pagination

Case Loading

Case Error

Case Success

Case Warning

Case Info

Case Help

Case Support

Case Contact

Case Feedback

Case Rating

Case Review

Case Approval

Case Denial

Case Cancellation

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Case Sort

Case Pagination

Case Loading

Case Error

Case Success

Case Warning

Case Info

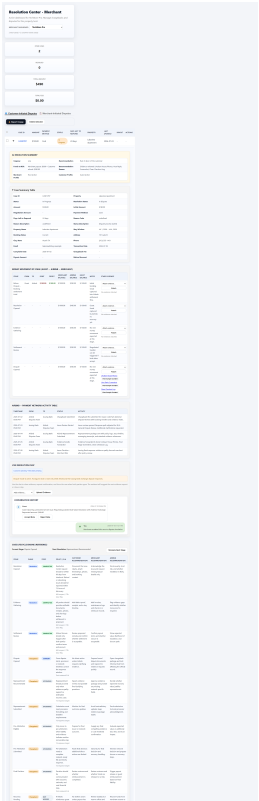
Case Help

Case Support

Case Contact

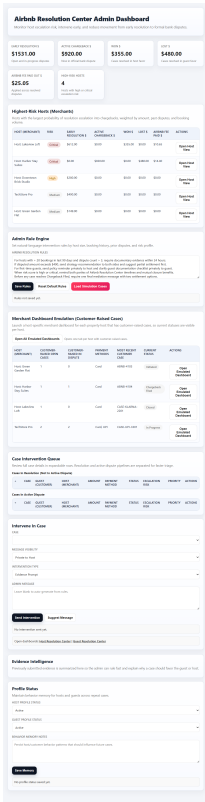
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Merchant Expanded Case: recommendation-led investigation and evidence trace



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Admin Expanded Case: intervention queue, settlement view, and lifecycle controls



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